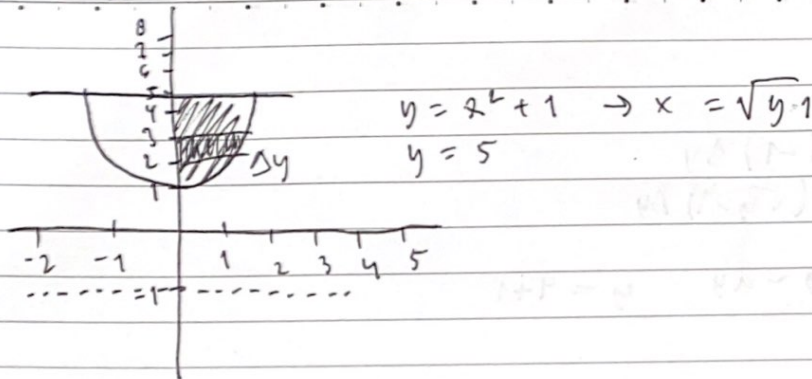


Tugas Minggu 11

No
Date



b. Luas daerahnya

$$\begin{aligned}\int_0^2 5 - x^2 - 1 dx &= \int_0^2 4 - x^2 dx \\ &= \left[4x - \frac{1}{3}x^3 \right]_0^2 \\ &= 8 - \frac{8}{3} = \frac{16}{3}\end{aligned}$$

c. 1. $\Delta V = \pi (\sqrt{y-1})^2 \Delta y$
 $V = \pi \int_1^5 y-1 dy$

$$\begin{aligned}&= \pi \left(\frac{1}{2}y^2 - y \right) \Big|_1^5 = \pi \left(\frac{25}{2} - 5 - \left(\frac{1}{2} - 1 \right) \right) \\ &= \pi (13 - 5) \\ &= 8\pi\end{aligned}$$

Cara kulit tabung

$$\begin{aligned}1. \Delta V &= 2\pi (5x^2 - 1) \times \Delta x \\ &= 2\pi (4x - x^3) \\ V &= 2\pi \left[2x^2 - \frac{1}{4}x^4 \right]_0^2 dx \\ &= 2\pi (8 - 4) \\ &= 8\pi\end{aligned}$$

Cara Cakram

$$\begin{aligned}2. \Delta V &= \pi ((5+2)^2 - (x^2 + 1 + 2)) \Delta x \\ V &= \pi \int_0^2 49 - (x^2 + 3) dx \\ &= \pi \int_0^2 46 - x^2 dx \\ &= \pi \left(46x^2 - \frac{1}{3}x^3 \right) \Big|_0^2 \\ &= \frac{268}{3}\pi\end{aligned}$$

Cara kulit Tabung

$$\text{C2 } r = y+2$$

$$h = \sqrt{y-1}$$

$$\Delta V = 2\pi (y+2) (\sqrt{y}-1) \Delta y$$

$$V = 2\pi \int_1^5 (y+2) (\sqrt{y}-1) dy$$

$$= y-1 = y \quad dy = dy \quad y = y+1$$

$$\frac{dy}{dy} = 1$$

$$2\pi \int_1^5 (y+3) (\sqrt{y}) dy$$

$$2\pi \int_1^5 u^{3/2} + 3u^{1/2} du$$

$$2\pi \left(\frac{2}{5} u^{5/2} + \frac{6}{3} u^{3/2} \right) \Big|_0^4$$

$$2\pi \left(\frac{2}{5} u^{5/2} + \frac{6}{3} 2 u^{3/2} \right) \Big|_0^4$$

$$= 2\pi \left(\frac{144}{5} \right)$$

$$= \frac{288}{5} \pi //$$